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Aiyshwarya aruchamy

Computer Vision Engineer



EDUCATION

Integrated M.Sc. in Data Science

Jul '19 - Jun '24

Amrita Vishwa Vidyapeetham

Coimbatore, Tamil Nadu

KEY SKILLS

Programming & Data: Python, SQL (PostgreSQL, MySQL), Java, Linux.

Machine Learning: Supervised & Unsupervised Learning, Predictive Modeling, Forecasting, Feature Engineering, EDA.

Generative AI & NLP: Large Language Models (LLMs), Transformers, SentenceTransformers, Retrieval-Augmented Generation (RAG), Prompt Engineering, LangChain, Hugging Face.

Computer Vision: OpenCV, YOLO, Faster R-CNN, FCOS, SSD, Vision Transformers, GANs.

Frameworks & Libraries: PyTorch, TensorFlow, Keras, Scikit-learn, Pandas, NumPy, Matplotlib.

Cloud & Data Systems: AWS (EC2, S3), Azure, GCP, Spark, Hadoop.

PROFESSIONAL EXPERIENCE

Technology Analyst / Engineer

Nov '25 - Jan '26

Infosys Ltd. – Client: American Express

Bengaluru,

- Built and validated a GenAI-ready hybrid search & retrieval system combining BM25 lexical search with semantic vector embeddings (FAISS) to support relevance ranking and prediction use cases over millions of records.
- Developed efficient, reusable Python pipelines for data ingestion, cleansing, embedding generation, and integration across structured and unstructured data sources.

Computer Vision Engineer

Nov '24 - Nov '25

Wastefull Insights

Vadodara, Gujarat

- Designed, trained, and deployed YOLO-based object detection models for real-time industrial vision use cases, achieving ~90% production accuracy.
- Built end-to-end real-time video analytics pipelines, including preprocessing, augmentation, inference, and post-processing.
- Optimized inference using TensorRT, reducing end-to-end latency by ~30%, enabling deployment under real-time constraints.
- Fine-tuned CNN-based detection models on custom annotated datasets, following full ML model development lifecycle.
- Collaborated with cross-functional teams to communicate model trade-offs, performance benchmarks, and deployment constraints.
- Led dataset annotation efforts and validation workflows to improve model robustness and consistency.

Data analysis/AIOT-Support Intern

Aug '23 - Jul '24

Altair Engineering India Private Limited

Bangalore, India

- Developed and evaluated object detection models (Faster R-CNN, FCOS, Vision Transformers) on large-scale image and video datasets.
- Built real-time defect detection systems, improving automated quality inspection accuracy.
- Supported GPU-accelerated deployment pipelines using CUDA and TensorRT.
- Contributed to video-based analytics pipelines, integrating vision models into production workflows.

INTERNSHIP

Data Analyst Intern

Jun '21 - Sep '21

Shape AI

- Developed time series forecasting models (ARIMA, Exponential Smoothing), improving forecast accuracy by 30%.

Aug '22

Teachnook

- Built CNN-based image classification models using transfer learning and augmentation.
- Focused on model interpretability and robustness, aligning with real-world deployment needs.

PROJECTS

Hybrid Search & Retrieval POC |(Nov 2025 - Dec 2025) *Python, SQL, OpenSearch (BM25), FAISS, SentenceTransformers, NLP, AWS*| [Github Link](#)
Designed and implemented a production-style multilingual hybrid search engine for Indian-language documents, combining lexical search (BM25 via OpenSearch) with semantic vector search (FAISS IVF-PQ).

Generative AI – Video-Based Sentiment Analysis |(Jul 2021 - Dec 2021) *Python, AWS, Speech-to-Text, NLP, LLMs* | [Github Link](#)
Built an end-to-end NLP pipeline using LLM-based speech-to-text and sentiment models to predict trends and extract insights from video data; deployed on AWS for scalable processing.

Video Retrieval-Augmented Generation (RAG) Project |(06/2021 - 10/2021) *Python, LangChain, RAG, Docker, Kubernetes, AWS* | [Github Link](#)
Designed a GenAI POC using LangChain and Hugging Face to enable enterprise document retrieval and summarization, containerized with Docker and orchestrated using Kubernetes.

Facial Emotion Detection(02/2023 - 07/2023) *Python, Tensorflow, Power BI*| [Github Link](#)
Developed a transfer learning pipeline for facial emotion detection, leveraging augmentation and visualization methods to improve adaptability and interpretability of recognition models.

Image Classification Extension(09/2023 - 11/2023) *Python, Keras, Rapid Miner* | [Github Link](#)
Created a custom image classification operator with augmentation techniques (rotation, scaling, flipping), achieving a 15% accuracy boost and integrating into RapidMiner workflows for enterprise-scale deployment.

Object Detection and Image Segmentation Extension(12/2023 - 03/2024) *Python, PyTorch, GANs, TensorRT, RapidMiner*| [Github Link](#)
Engineered real-time detection and segmentation models (Faster R-CNN, SSD, GANs) optimized for edge computing using TensorRT, reducing latency while maintaining high accuracy.

Synthetic Waste Image Generation(07/2024 - 08/2024) *Python, OpenCV, PyTorch,Docker, AWS S3*| [Github Link](#)
Automated a synthetic data generation pipeline for waste images, containerized with Docker and integrated with AWS S3, enabling large-scale dataset augmentation for training deep learning models.

Annotation Transfer Tool(10/2024 - 11/2024) *Python, VGG, CNN, ResNet, PCA, SVD, Cosine Similarity, Euclidean Distance, CVAT* | [Github Link](#)
Built an automated annotation transfer tool using VGG/CNN feature extraction and similarity measures (PCA, SVD, Cosine Distance), reducing manual labeling effort by over 50% with CVAT validation support.

Raccoon Network Analysis(06/2022 - 08/2022) *Python, Pandas, NetworkX, Matplotlib, Seaborn* | [Github Link](#)
Modeled zoonotic disease spread in raccoon populations using network analytics, applying clustering and centrality metrics to identify high-risk transmission nodes for targeted intervention.

Bus Reservation and Ticketing System(02/2021 - 04/2021) *Java, MySQL* | [Github Link](#)
Developed a Java-based reservation and billing system with discount logic, improving ticketing efficiency and passenger management workflows.

CERTIFICATES

- Web Development Using Java Script | Shape AI Bootcamp | May '21
- Basic Web Development using HTML,CSS,JS| Devtown | Jan'22 - Mar '22
- Machine Learning: Python & R in Data Science | Udemy | Dec '23 - Mar '24
- AWS Cloud Practitioner Essentials | Coursera | Jun'24 - Aug'24